# Capstone Build

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Verified Study Buddy

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WORKBOOK · 1 OF 21

## Recall and choose your starting point

> Record your starting explanation for Capstone Build before opening notes. Show any later correction separately.

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

### Capstone Design

Without opening the old lesson, explain one key idea from Capstone Design. Give an example and a mistake that the idea helps you avoid.

> **My remembered idea, example and correction**

### Safety Boundaries

Without opening the old lesson, explain one key idea from Safety Boundaries. Give an example and a mistake that the idea helps you avoid.

**My remembered idea, example and correction**

### Privacy by Design

Without opening the old lesson, explain one key idea from Privacy by Design. Give an example and a mistake that the idea helps you avoid.

**My remembered idea, example and correction**

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## Practise with fading help

> Complete the supported Capstone Build trace and record the first step you can explain without help.

Use W2: Trace the worked case for Capstone grounded assistant. Record the answer once; the lesson and workbook refer to the same attempt.

| Given inputs | Procedure I chose | Middle steps | Result and check |
|---|---|---|---|
| | | | |

**The step I completed without help, and the hint I needed**

WORKBOOK · 3 OF 21

## Individual reasoning check

> Check the proposed Capstone Build explanation against the supplied facts before discussing it.

Use the worked case for Capstone grounded assistant. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

**My explanation before partner discussion**

WORKBOOK · 4 OF 21

## Independent transfer case

> Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

### Independent case: Changed independent case

An experiment note records four fictional seeds observed for six days but supplies no growth results. A generated draft says all four sprouted and the method always works in any soil. Repair it.

**My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

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WORKBOOK · 5 OF 21

## Your learning target

> Connect the Capstone Build target to a trace and a justified conclusion.

> A prototype cites note A but its answer adds a detail found nowhere in A. What should the final validation stage check and do?

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WORKBOOK · 6 OF 21

## Solve the given case

> Solve the supplied Capstone Build case and show the deciding evidence.

A pipeline retrieves passages, drafts and displays without checking citations.

What step is missing?

Show your trace, calculation or **evidence**

**Write or draw your reasoning and evidence.**

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WORKBOOK · 7 OF 21

## Check a changed case

Apply the procedure to the changed case and record why the conclusion follows.

The interface looks complete, but its answer button returns the same saved sentence for every question. What capability is still missing, and how can you expose it?

My independent answer and reason

**Write or draw your reasoning and evidence.**

WORKBOOK · 8 OF 21

## Design a revealing test

> Specify a changed input, expected result and observation separately.

Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

| Test record | My evidence |
| --- | --- |
| Changed input or condition | |
| Expected answer or fallback | |
| Observed result | |
| Why the result occurred | |

WORKBOOK · 9 OF 21

## Project checkpoint

> Justify what Capstone Build adds to Verified Study Buddy and keep the evidence.

Assemble interface, retrieval, prompt, model/tool, validation, **evidence** display, and fallback.

The project change and **evidence** I will keep

**Write or draw your reasoning and evidence.**

WORKBOOK · 10 OF 21

## Self check

> Judge explanation, evidence, transfer and limits separately; attach the work behind your judgement.

- I can explain the mechanism.
- I can show evidence.
- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

WORKBOOK · 11 OF 21

## Transfer practice from the original programme

> Use this earlier case as transfer practice. A case already discussed with feedback is not fresh assessment evidence.

A prototype cites note A but its answer adds a detail found nowhere in A. What should the final validation stage check and do?

**My independently reasoned answer**

Record support used: none, hint or teacher explanation.

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## W1 Explain the mechanism

Explain the mechanism. Apply the supplied Capstone Build facts and keep your reasoning.

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

**My reasoning and evidence**

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## W2 Trace the worked case

> Trace the worked case. Apply the supplied Capstone Build facts and keep your reasoning.

Request room B; retrieved approval names B but register names C. Draft gives capacity 10 from C. Show the intermediate steps and give the conclusion with its scope.

**My reasoning and evidence**



## W3 Audit the limit

> Audit the limit. Apply the supplied Capstone Build facts and keep your reasoning.

A report omits this qualification: A correct happy path does not establish fallback, access control or side-effect safety. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

**My reasoning and evidence**

## W4 Apply a new case independently

> Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Apply a new case independently. Apply the supplied Capstone Build facts and keep your reasoning.

Request room J; approval and register both J capacity 0. Draft is well-formed with J/0 and correct sources. Trace stages and distinguish zero from missing.

> **My reasoning and evidence**

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## W5 Design a revealing follow-up

> Design a revealing follow-up. Apply the supplied Capstone Build facts and keep your reasoning.

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

**My reasoning and evidence**

WORKBOOK · 17 OF 21

## Feedback, return check and learning record

> Record the first repaired step and a different case to revisit after a delay.

| Evidence | My record |
| --- | --- |
| Worked alone / hint / model | |
| First step I repaired and why | |
| New case checked later; date and result | |
| What I can now explain without notes | |
| One remaining question | |

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

WORKBOOK · 18 OF 21

## Pictorial case practice: Capstone Build

> Build a claim checker for a draft summary of a fictional class investigation. Explain the

> labelled step before checking the changed condition.

Build a claim checker for a draft summary of a fictional class investigation.

Approved record E2: three cups were observed for four days. No measured water loss is provided. Draft says “three cups were observed for four days and all lost half their water”.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

AI ACADEMY / AI-3/34

## Capstone Build

**Start with the supplied facts**

Follow each arrow. Ask which fact allows the next step.

### 1. Extract claims

Separate cup count, observation duration and the claimed water loss.

↓

### 2. Match evidence

E2 supports three cups and four days; it supplies no water-loss result.

↓

### 3. Repair draft

Retain the supported procedure facts; state that water-loss results were not supplied.

**CHANGE ONE THING**

E3 supplies losses of 1, 2 and 3 mL.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

What additional numbers would a half-water claim require? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

**My picture-based explanation and changed-case prediction**

### Record the help used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: For your Verified Study Buddy, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

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## Project portfolio milestone

> Keep this week's input, deciding step, result and limit together in the Verified Study Buddy record.

Project: Verified Study Buddy

Current milestone: Present and defend. Demonstrate the system, explain one failure and justify the next improvement with **evidence**.

For your Verified Study Buddy, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

| Evidence to keep | My record |
|---|---|
| This week's input and deciding step | |
| Prediction and actual result (or labelled paper trace) | |
| One change since the previous checkpoint and why | |

| Evidence to keep | My record |
| --- | --- |
| One error or limit and the next test | |

**Explain what this week adds to your project. Reuse last week's record rather than starting again.**

WORKBOOK · 20 OF 21

## Supported practice and learning evidence

> Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Build a claim checker for a draft summary of a fictional class investigation.

### Practice 1: explain the deciding evidence

Which labelled step matches this detail? Retain the supported procedure facts; state that water-loss results were not supplied.

1. Extract claims

2. Match **evidence**

3. Repair draft

**My selected conclusion, deciding fact and explanation**

### Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

### Feedback · compare after trying

After your attempt, compare with the supported explanation: Repair draft Retain the supported procedure facts; state that water-loss results were not supplied. Read the picture in order.

Retain the supported procedure facts; state that water-loss results were not supplied. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

**Practice 2: explain the deciding evidence**

### Changed case · predict first

Changed-case practice: E3 supplies losses of 1, 2 and 3 mL. Which conclusion is supported by the supplied facts?

1. We can decide the result without examining the changed **input** or the procedure.

2. Report those individual losses, not “half” unless starting volumes are also available.

3. This one example guarantees correct results for every future **input**.

**My selected conclusion, deciding fact and explanation**

### Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

### Feedback · compare after trying

After your attempt, compare with the supported explanation: Report those individual losses, not “half” unless starting volumes are also available. Absolute losses alone do not establish each cup’s fractional loss.

### Repair a missing building block

Review Capstone Design (ai-3/33). Goal: Explain the **mechanism** and terminology of capstone requirements and corpus. Short practice: What should the project plan record before the new flower test?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Safety Boundaries (ai-3/31). Goal: Explain the **mechanism** and terminology of untrusted **input** boundaries. Short practice: If AB7 passes the pattern, may it access teacher-only material?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Privacy by Design (ai-3/28). Goal: Explain the **mechanism** and terminology of data retention and deletion. Short practice: Which IDs and locations must appear in the deletion **evidence**?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

### Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

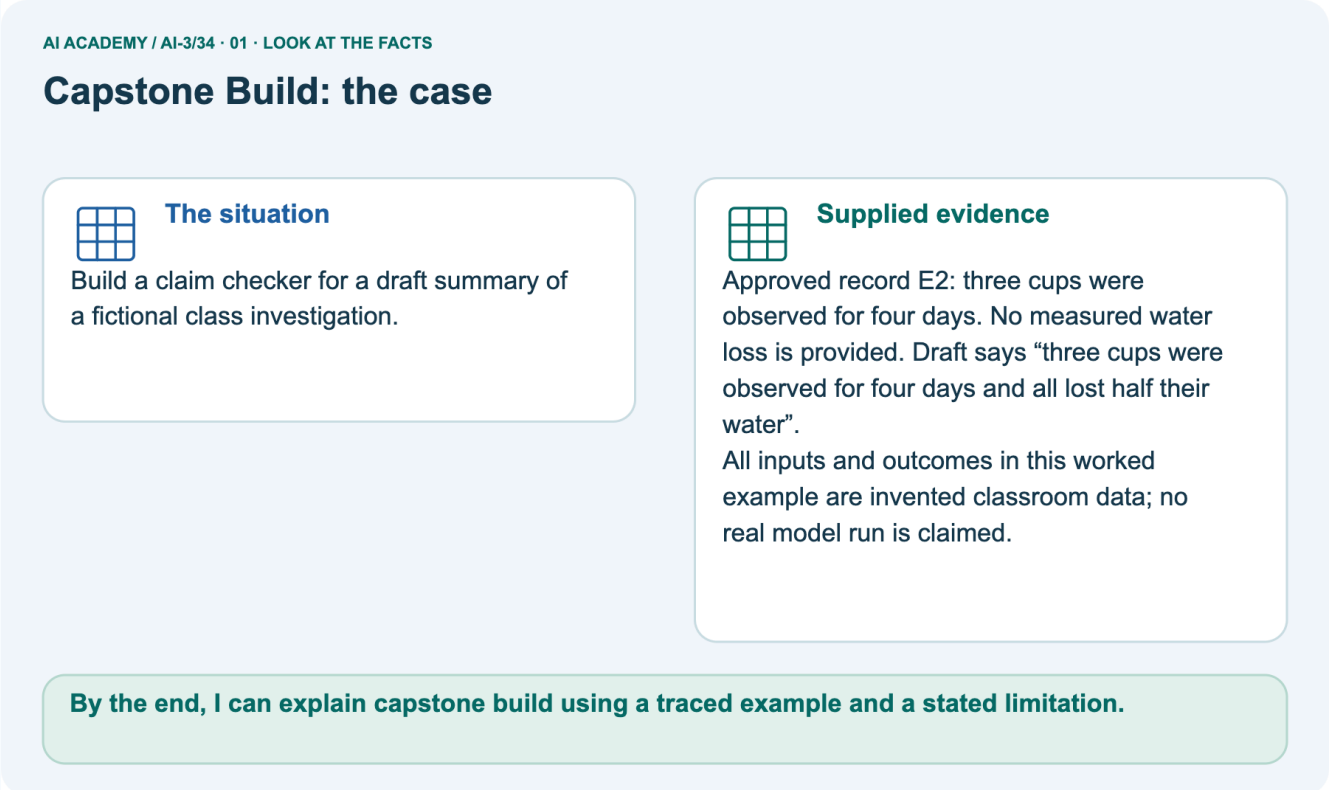
**What I repaired, the evidence I kept and the next test**

WORKBOOK · 21 OF 21

# Visual concept journey

> Explain each Capstone Build picture from its labels, then record a relevant limit.

### 01. Capstone Build: the case




Follow the labelled picture with the matching explanation. Open the vector diagram

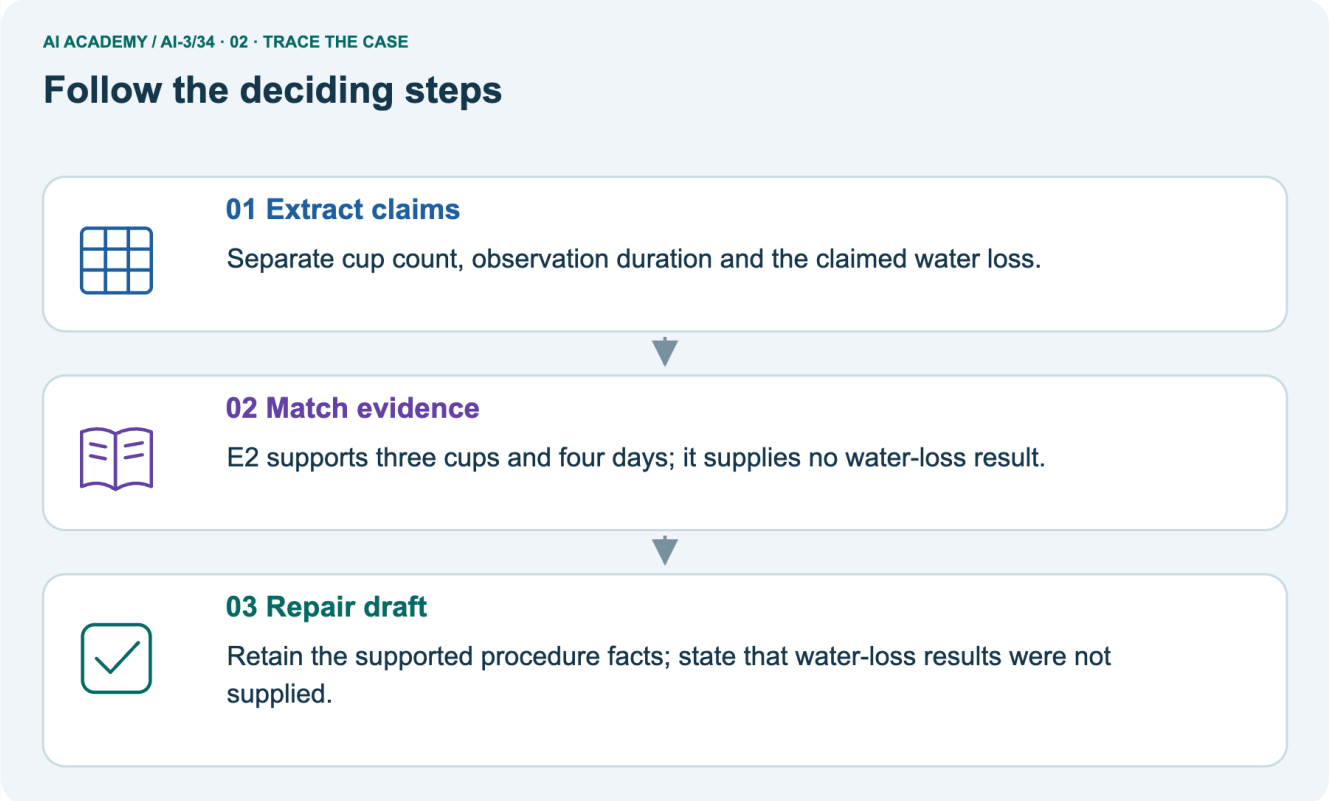
Build a claim checker for a draft summary of a fictional class investigation. Supplied facts: Approved record E2: three cups were observed for four days. No measured water loss is provided.

Draft says “three cups were observed for four days and all lost half their water”. All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

> Point to the given fact that matters most. Explain why it matters.

**My picture explanation, deciding evidence and one limit**

## 02. Follow the deciding steps




Follow the labelled picture with the matching explanation. Open the vector diagram

Extract claims: Separate cup count, **observation** duration and the claimed water loss. → Match **evidence**: E2 supports three cups and four days; it supplies no water-loss result. → Repair draft: Retain the supported procedure facts; state that water-loss results were not supplied.

Point to each numbered stage and explain which recorded **input** it uses.

**My picture explanation, deciding evidence and one limit**

### 03. Capstone grounded assistant — evidence trace

AI ACADEMY / AI-3/34 · 03 · INSPECT THE LESSON MODEL

**Capstone grounded assistant — evidence trace**

| Input and scope | Check or calculation | Conclusion and limit |
| --- | --- | --- |
| Request room B; retrieved approval names B but register names C. Draft gives capacity 10 from C. | A supported response requires current matching approval and register records. Any retrieval or validation failure gives needs_review with null claims. The classroom implementation performs no booking or external action. | Same-room support fails before or during semantic validation. Return needs_review with null claims; the draft capacity is unsupported for B. |

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Follow the labelled picture with the matching explanation. Open the vector diagram

Capstone grounded assistant — **evidence** trace. Request room B; retrieved approval names B but register names C. Draft gives capacity 10 from C. | A supported response requires current matching approval and register records.

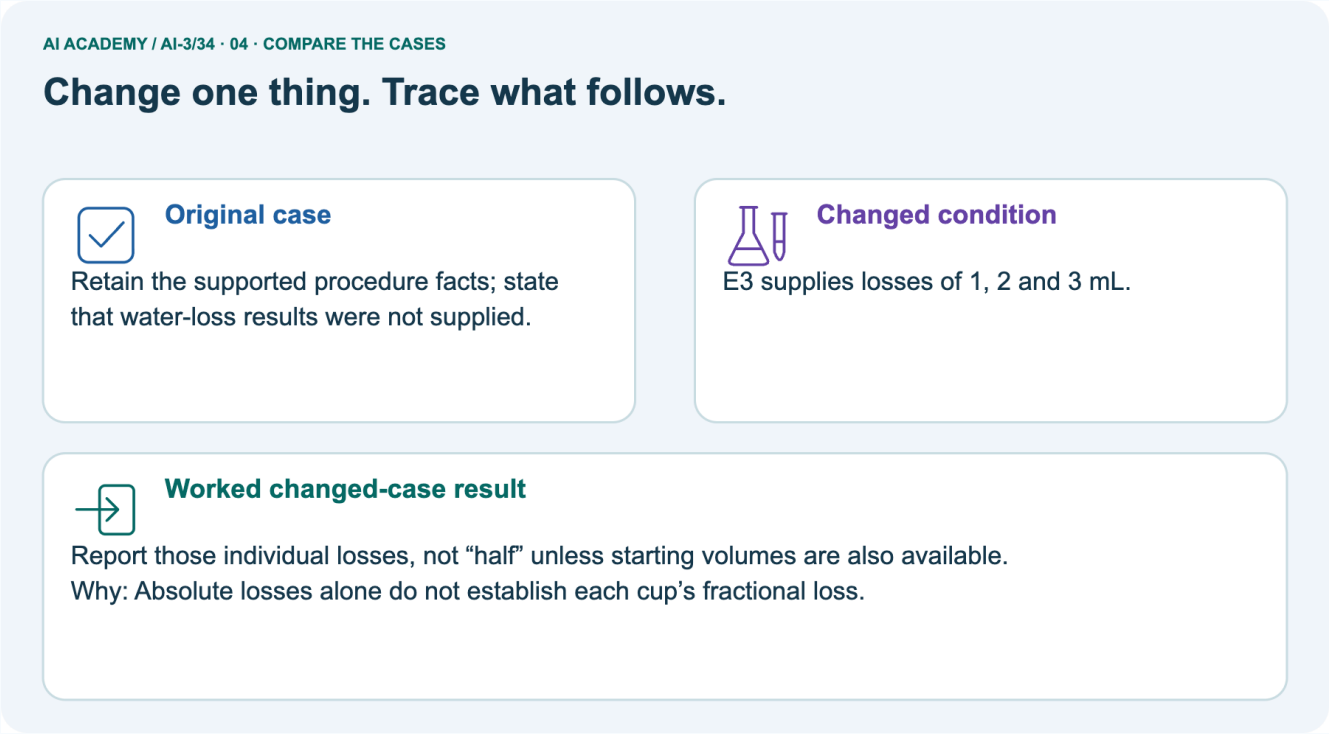
Any retrieval or validation failure gives needs_review with null claims. The classroom implementation performs no booking or external action. | Same-room support fails before or during semantic validation.

Return needs_review with null claims; the draft capacity is unsupported for B.

> Read the labels and values. Explain the relationship the model shows.

**My picture explanation, deciding evidence and one limit**

## 04. Change one thing. Trace what follows.




Follow the labelled picture with the matching explanation. Open the vector diagram

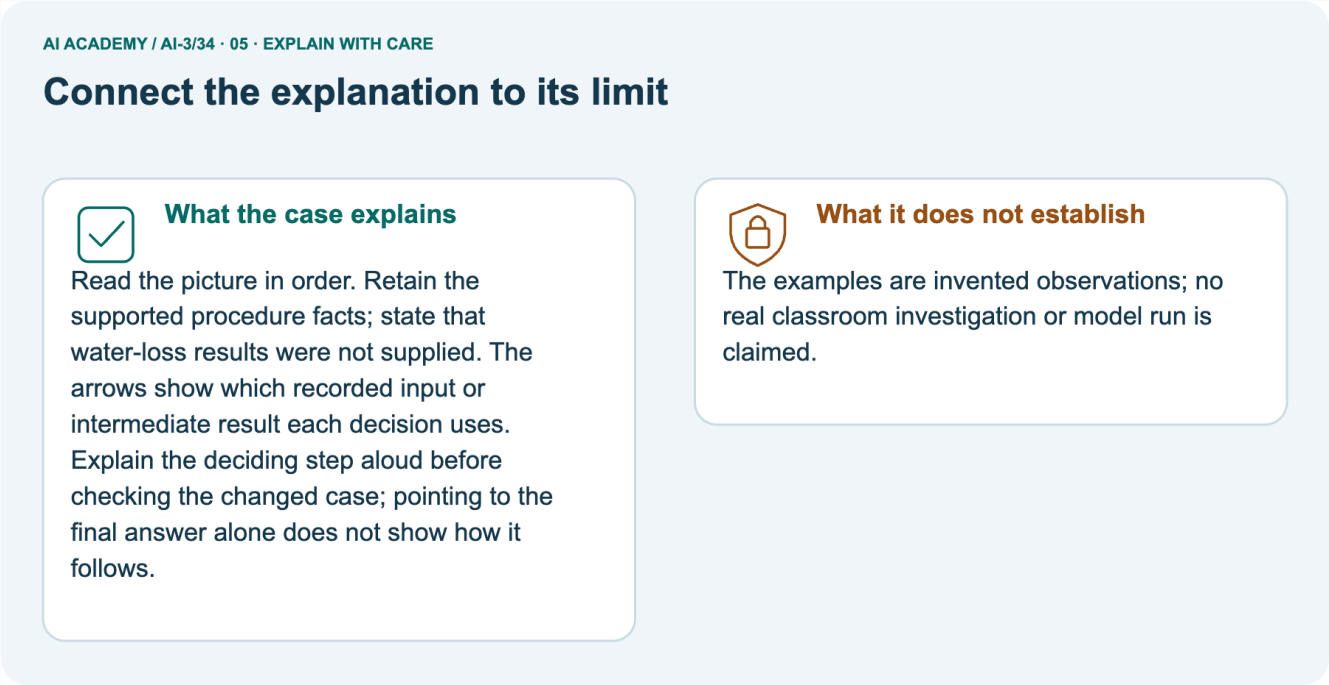
Original case: Retain the supported procedure facts; state that water-loss results were not supplied. Changed condition: E3 supplies losses of 1, 2 and 3 mL.

Expected result: Report those individual losses, not “half” unless starting volumes are also available. Reason: Absolute losses alone do not establish each cup’s fractional loss.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

### My picture explanation, deciding evidence and one limit

## 05. Connect the explanation to its limit




Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture in order. Retain the supported procedure facts; state that water-loss results were not supplied. The arrows show which recorded **input** or intermediate result each decision uses.

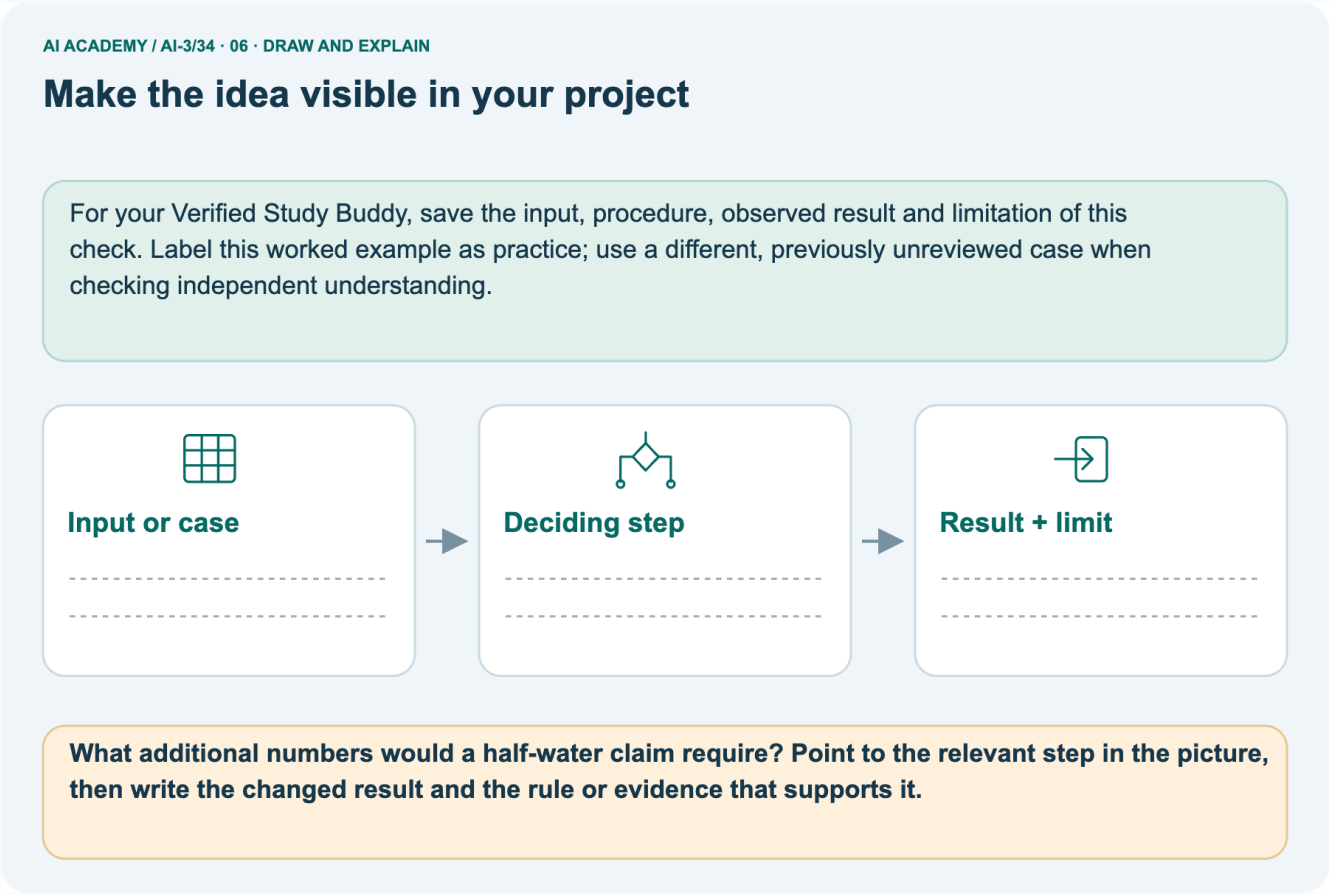
Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Limit: The examples are invented **observation**s; no real classroom investigation or model run is claimed.

> Explain the case, then point to a limit on the claim.

**My picture explanation, deciding evidence and one limit**

## 06. Make the idea visible in your project




Follow the labelled picture with the matching explanation. Open the vector diagram

For your Verified Study Buddy, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: What additional numbers would a half-water claim require? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> Draw your project trace and justify the deciding step using **evidence**.

### My picture explanation, deciding evidence and one limit

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.